VIZ 02

DATA AND TASK CLASSIFICATION

LADISLAV ČMOLÍK

Department of Computer Graphics and Interaction
Faculty of Electrical Engineering at CTU in Prague

DCGI

Introduction

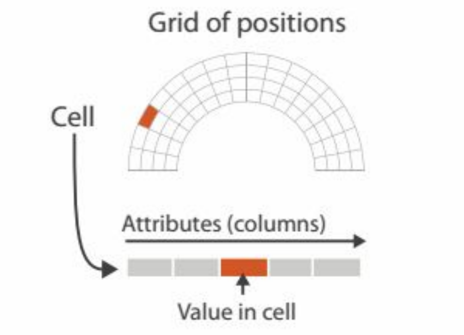
- There is a bewildering range of visualization techniques
  - How do we know which to choose?
- First step is to classify the data and tasks and associate different techniques with data types and task types
  - Data types - what we are visualizing
  - Tasks types - why we are visualizing the data
  - Visualization techniques - how we accomplish certain task with certain data

Dataset types

Spatial data

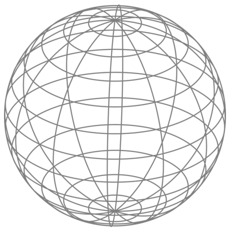
Spatial fields

- Fields
  - Grids
    - Positions
    - Cells
  - Attributes



Geometry

- Meshes
  - Vertices
  - Edges
  - Faces
  - Typically, no attributes



Scientific visualization techniques are used

Dataset types

Abstract data

Tabular data

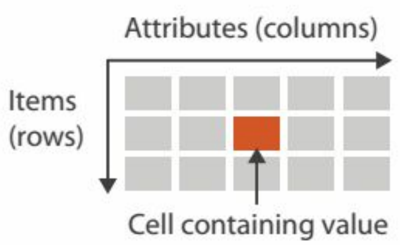
- Tables
  - Items
  - Attributes

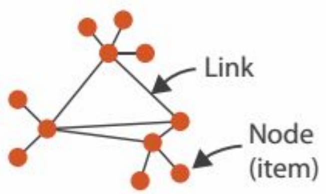
Relational data

- Networks
  - Items (nodes)
  - Links
  - Attributes

Text

- Document
  - Sections
  - Words
- Collection of documents
  - Source code





Information visualization techniques are used

Data enrichment makes no sense

Dataset types

Mixing spatial and abstract data

- Geographical data
  - We have geometry
    - Points of interest
    - Polylines, curves - streets, rivers
    - Areas - Country, district, parcel
  - To the geometry are attached abstract data
    - Text - name of street or district
    - Population in the area
    - Etc.

Time varying data

- Attributes and/or their structure (topology) change in time
- Structure can change for
  - Spatial data
    - Topology of grid
    - Topology of mesh
  - Abstract data
    - Topology of network
    - Structure of document

Attribute Types

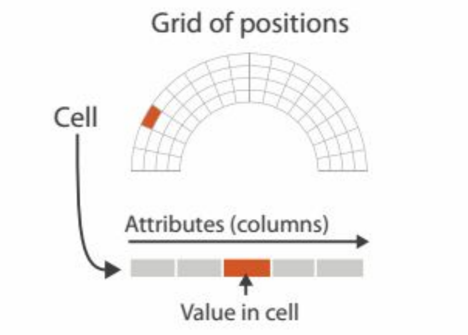
- Nominal/Categorical
  - No inherent order (in the sense of certain quantity)
  - People, Companies, Cities, Types of diseases, ...
- Ordered
  - Ordinal
    - Ordered, but not at measurable intervals
    - XS, S, M, L, XL, XXL, XXXL
    - Mon, Tue, Wed, Thu …
  - Quantitative
    - Ordered, arithmetic operations are possible
    - Discrete: Integers
    - Continuous: Floats

Attribute Types

- Ordering direction
  - Sequential - the values of the attribute range from minimum to maximum value
    - E.g., mountain height measured from sea level
  - Diverging - the values of the attribute can be split into two groups that diverge in positive and negative direction from a common zero point
    - E.g., Full elevation dataset containing mountain heights and depths of ocean floor is diverging
  - Cyclic - the values of the attribute wrap around to the starting point
    - E.g., Hour of the day, Day of the week

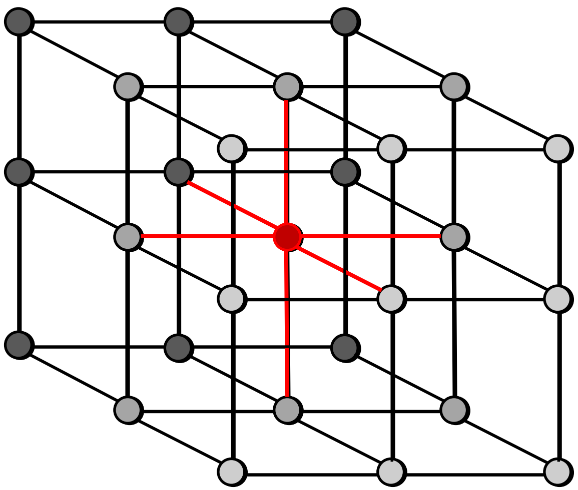
SPATIAL FIELDS

Fields – Attributes in grids



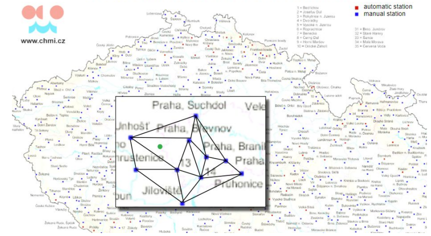
Spatial fields

- Spatial fields have a structure
  - The structure is represented with a grid
- Grid is similar to a geometry (3D mesh), but it may not be a surface
- Grid consists of
  - Positions (vertices),
  - Connectivity (edges) and
  - Cells
- Attributes
  - Both positions and cells may contain values of several different attributes
  - Temperature, pressure, velocity, …



Back to the Simple Example

Rainfall measurement stations ČHMÚ



- The underlying field is a function F(x, y)
  - F(x, y) represents the rainfall and is the **DEPENDENT** variable
  - x and y represent spatial coordinates and are **INDEPENDENT** variables
- x and y are positions on which we measured the data
- The grid is given by triangulation of the samples
- F(x, y) is the attribute (the data)
- It is a **two dimensional scalar** field
  - The positions x, y are 2D
  - the attribute F(x, y) is scalar

General Field Classification Scheme

- The underlying field can be regarded as a function of many variables: say

$$\mathbf{F}(\mathbf{x})$$

where $\mathbf{x}$ and $\mathbf{F}(\mathbf{x})$ are both vectors:

$$\mathbf{x} = (x_1, x_2, \ldots x_n)$$
$$\mathbf{F}(\mathbf{x}) = (F_1, F_2, \ldots F_m)$$

- The dimension is n
- The dependent variable can be scalar (m=1) or vector (giving us direction and size) (m>1)

A Simple Notation

- This leads to a simple classification of data as:

$$E_n^{S/V}$$

- So, the simple example is of type:

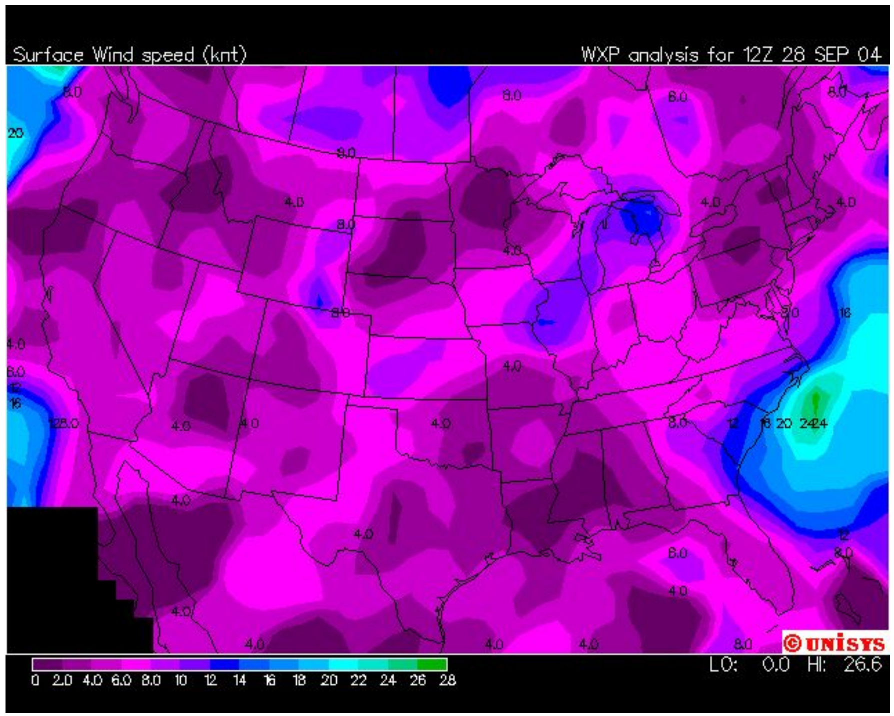
$$E_2^{S}$$

- Flow within a volume can be classified as:

$$E_3^{V_3}$$

Example - E^S_2

- Here we see a contour map of wind speed over the USA (28-Sep-04)



http://weather.unisys.com/surface/

Example - E^S_3

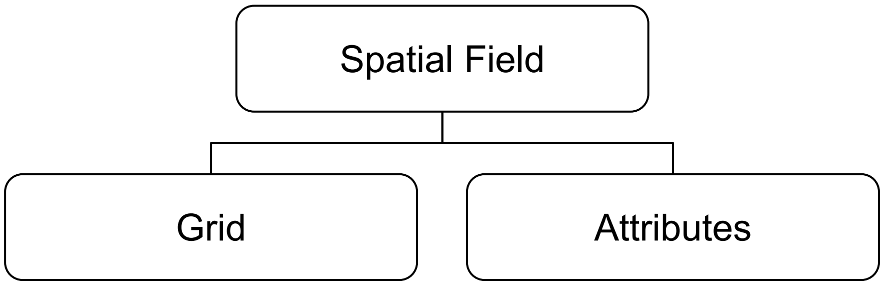
- A mouse skull (CT) visualized with volume rendering

Volume rendering of a mouse skull (CT) using shear warp algorithm, Christian Lackas (delta@lackas.net).

- As dimension increases visualization becomes more complex
- If the dependent variable is a vector, then the visualization is even more complex

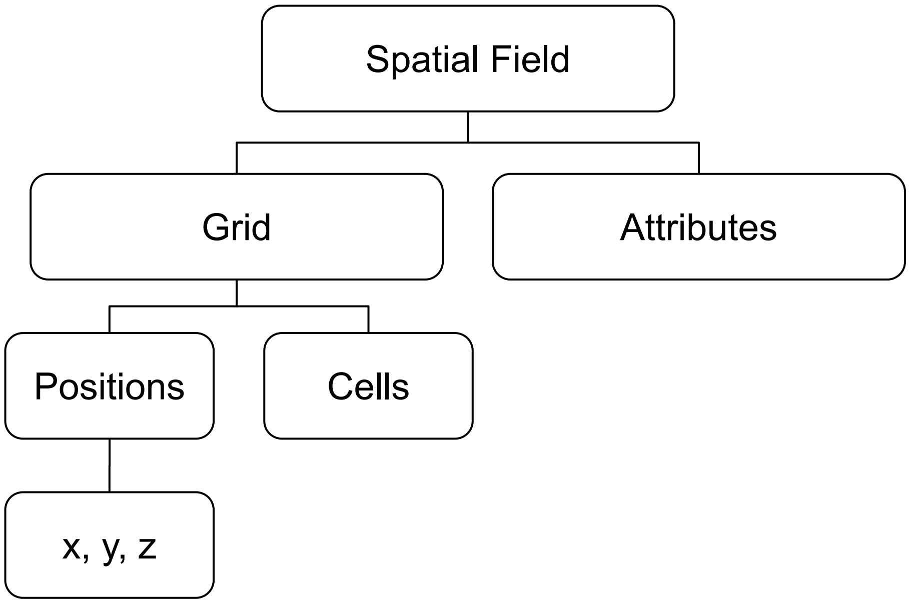
Spatial field

- Spatial field consists of:
  - **Grid:** the way the data samples relate to each other
  - **Attributes:** the data samples themselves

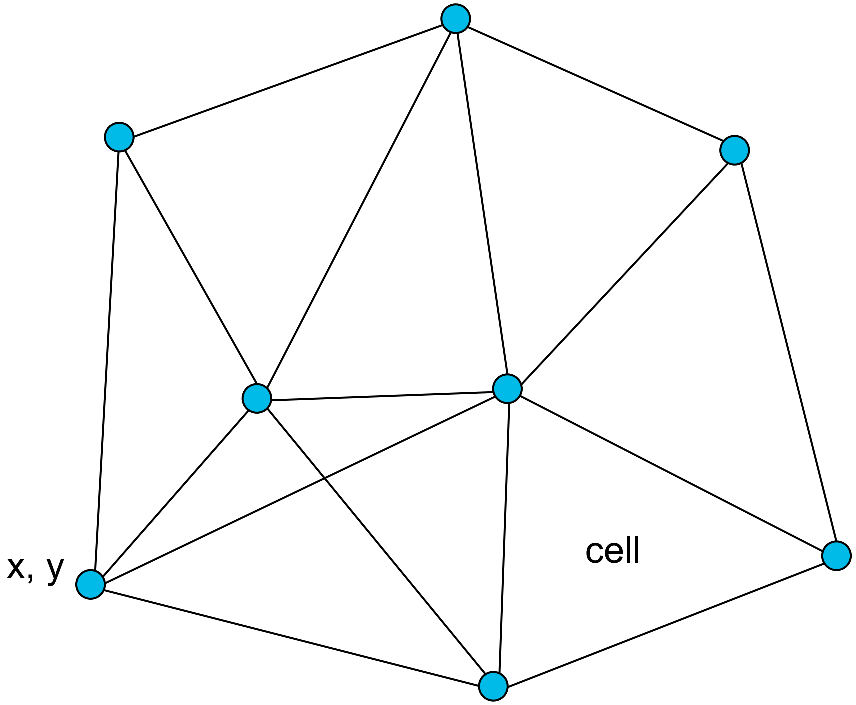


Grid

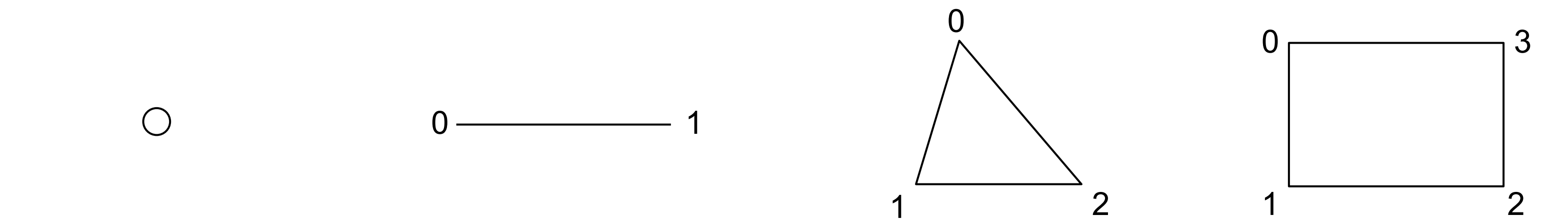
- Data samples are **positions** in space where we have measured or simulated the data
- Data samples are organized into a **grid**
- The grid and its **cells** is given by the **positions** and the **connectivity**
- The grid says from which samples to **interpolate**



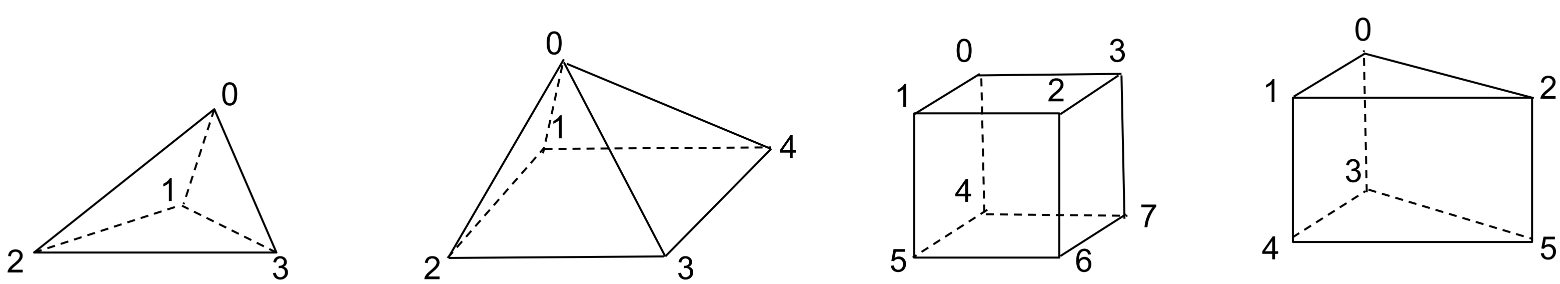
2D Grid Example



Cell types: 0D - 3D



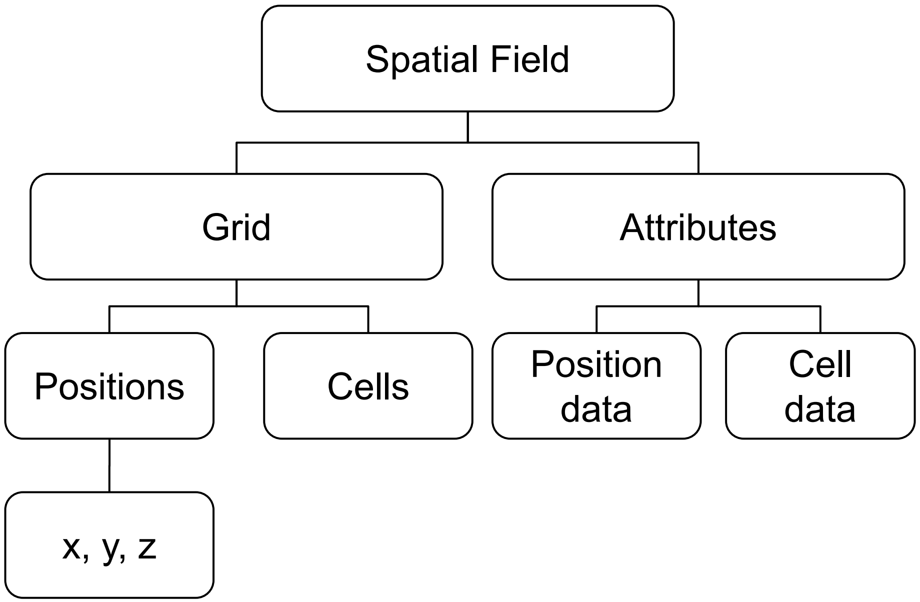
point line triangle rectangle

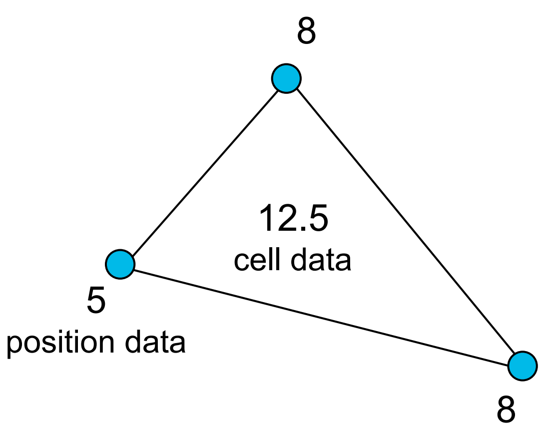


tetrahedron pyramid cube prism

Attributes

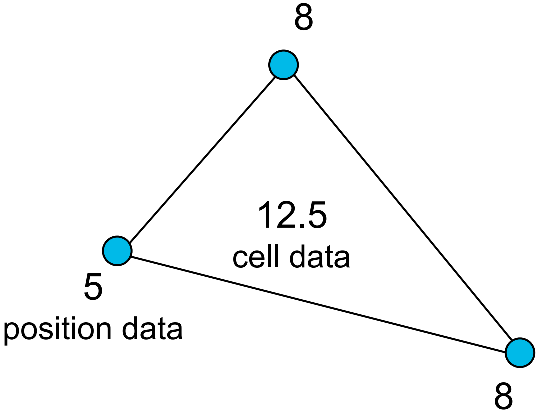
- Attributes are **values** of the measured/simulated data
- Attributes can be attached to **positions** and/or **cells**





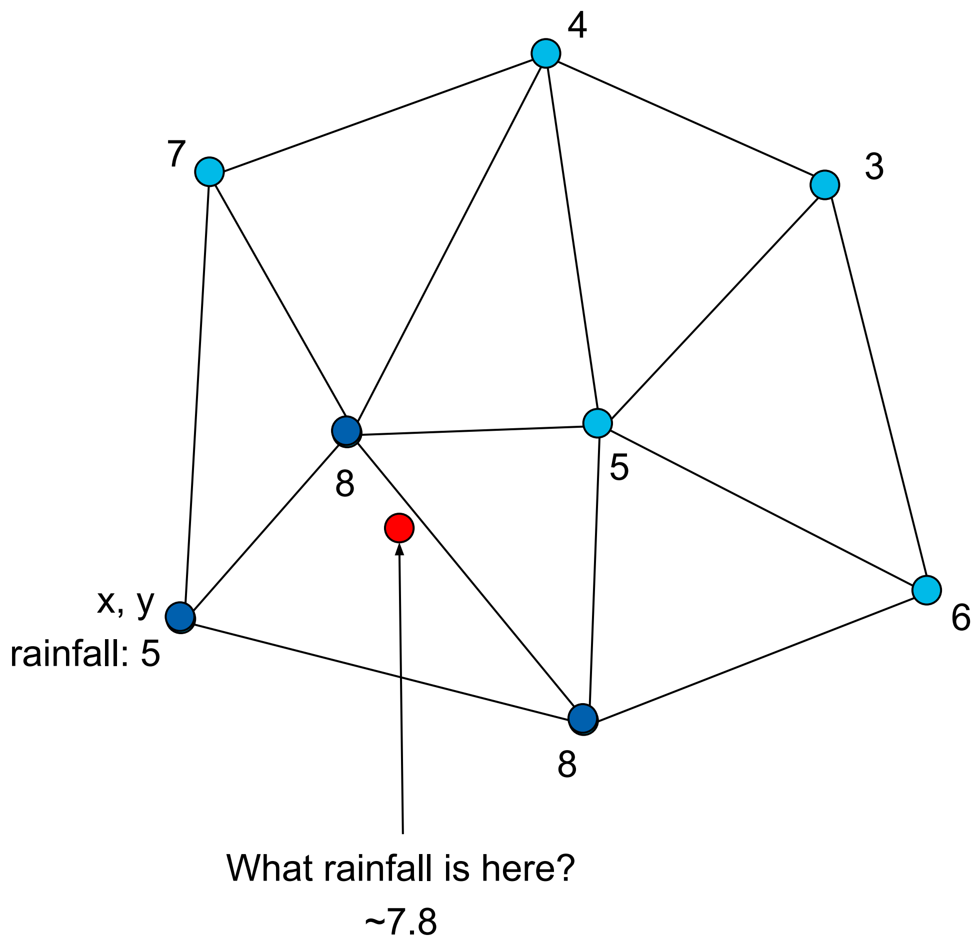
Typical types of attributes

- Scalar
  - temperature, pressure, velocity
- Vector
  - magnetic field, speed
- Tensor
  - stress, strain
- Multivariate
  - weather characteristics, water quality factors
  - E.g., temperature and wind direction and speed
  - We will see multivariate data as several fields (e.g., scalar and vector field)



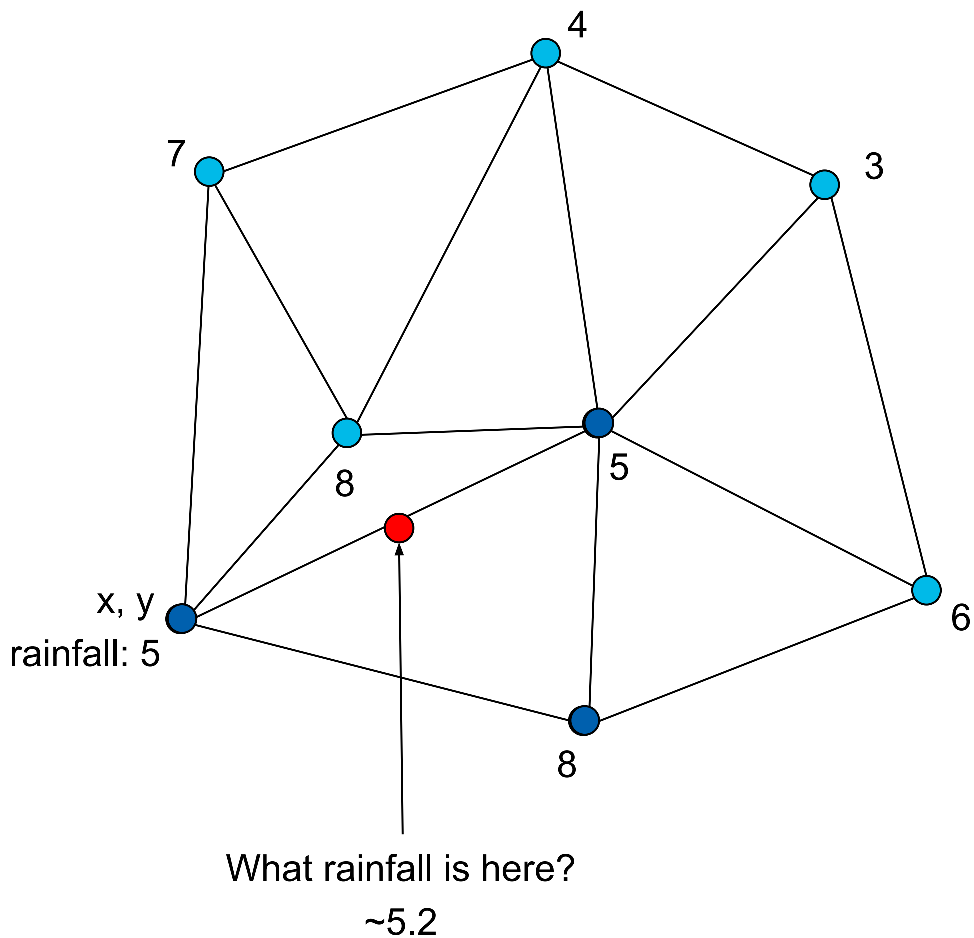
2D Grid Example

- The grid says from which position samples to **interpolate**

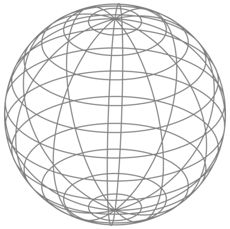


2D Grid Example

- Connectivity of the grid can have big influence on interpolated values



GEOMETRY



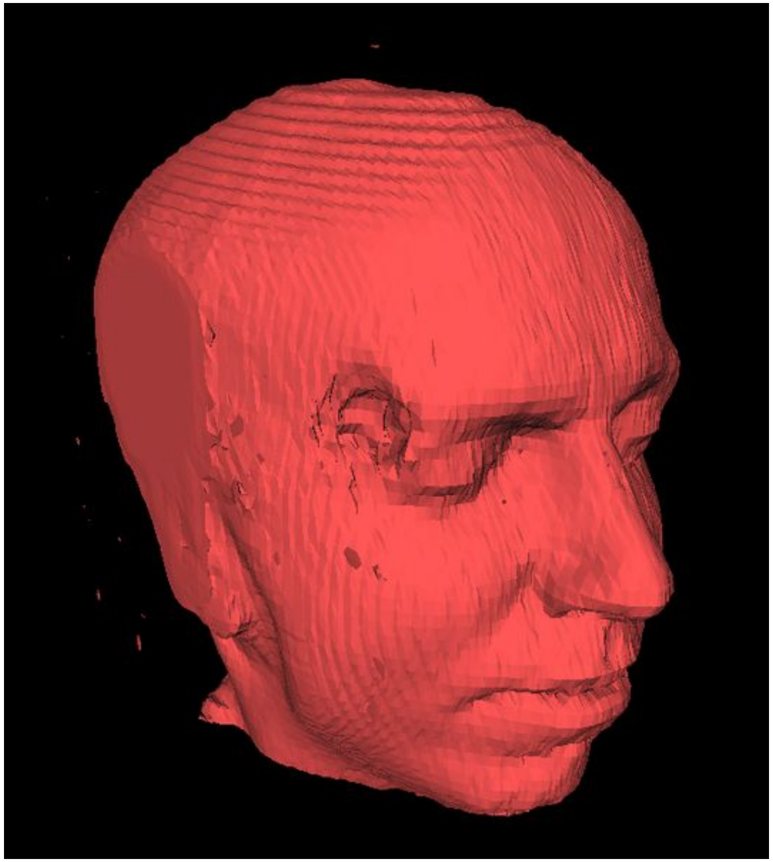
Field vs. Geometry

- In geometric modeling the geometry is used for object description
- Line, Area or Surface can be described by means of geometry
- Geometry consists of
  - Vertices - position
  - Edges - topology
  - Faces - equivalent of cells
  - There are no data attributes in vertices and faces
- Acquisition, modeling, and rendering of geometry is the subject of *computer graphics*
- In visualization geometry may be result of certain operations
  - Contouring, stream ribbon, etc.
- In geographical data, geometry describes the shape of a complex feature
  - River, country, state

3D geometry

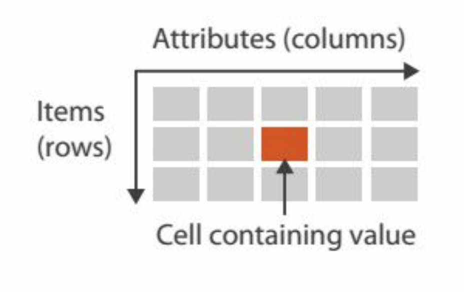
E_3^S

- Extracted from MRI dataset (3D scalar field) with contouring (Marching cubes)



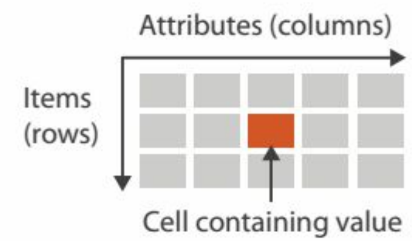
TABULAR DATA

Items and their attributes



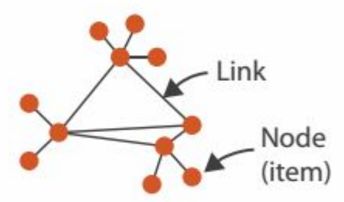
Tabular data

- Items
  - Rows of the table
  - Each row represent one item
- Attributes
  - Columns of the table
  - Each column is one attribute
- Cell
  - Intersection of
    - Certain row (Item)
    - Certain column (attribute)
    - Contain the **value** of the **attribute** for the **item**
- Data are not spatiotemporal
  - The data typically do not contain information about position/time where/when they were measured
  - Data enrichment is not applicable
    - Makes no sense to interpolate between several filled in questionnaires
- Information visualization techniques are used



RELATIONAL DATA

Nodes, links and their attributes



Relational data

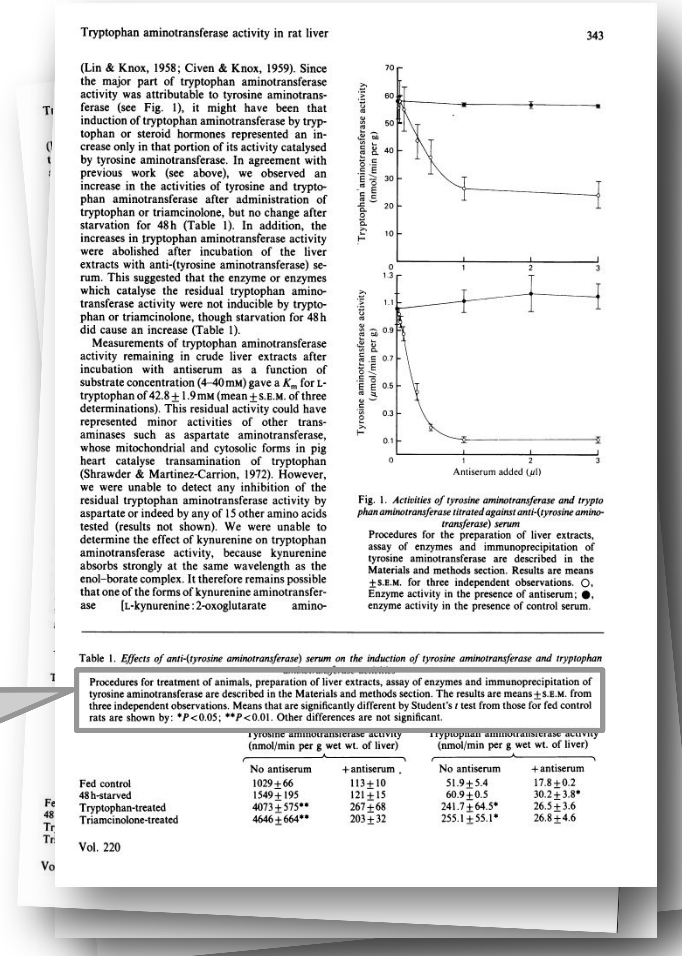
- Similar to tabular data, but contain **relations** (**links**) between **nodes** (items) in one or more tables
  - To express hierarchy (Tree) or arbitrary relations (Network)
- Attributes
  - Both nodes and links can store values of several different attributes
- Data can be spatiotemporal
  - E.g., Nodes are cities and links are airline flights
- Information visualization techniques are used

TEXT

Loosely structured data

Text and Document Data

- We distinguish between visualization of
  - Text itself (document)
  - Collections of
    text documents (corpus)
  - Streams (e.g., Twitter)



Tryptophan aminotransferase activity in rat liver 343

(Lin & Knox, 1958; Civen & Knox, 1959). Since the major part of tryptophan aminotransferase activity was attributable to tyrosine aminotransferase (see Fig. 1), it might have been that induction of tryptophan aminotransferase by tryptophan or steroid hormones represented an increase only in that portion of its activity catalysed by tyrosine aminotransferase. In agreement with previous work (see above), we observed an increase in the activities of tyrosine and tryptophan aminotransferase after administration of tryptophan or triamcinolone, but no change after starvation for 48h (Table 1). In addition, the increases in tryptophan aminotransferase activity were abolished after incubation of the liver extracts with anti-(tyrosine aminotransferase) serum. This suggested that the enzyme or enzymes which catalyse the residual tryptophan aminotransferase activity were not inducible by tryptophan or triamcinolone, though starvation for 48h did cause an increase (Table 1).

Measurements of tryptophan aminotransferase activity remaining in crude liver extracts after incubation with antiserum as a function of substrate concentration (4–40mM) gave a K_m for L-tryptophan of 42.8±1.9mM (mean±S.E.M. of three determinations). This residual activity could have represented minor activities of other transaminases such as aspartate aminotransferase, whose mitochondrial and cytosolic forms in pig heart catalyse transamination of tryptophan (Shrawder & Martinez-Carrion, 1972). However, we were unable to detect any inhibition of the residual tryptophan aminotransferase activity by aspartate or indeed by any of 15 other amino acids tested (results not shown). We were unable to determine the effect of kynurenine on tryptophan aminotransferase activity, because kynurenine absorbs strongly at the same wavelength as the enol–borate complex. It therefore remains possible that one of the forms of kynurenine aminotransferase [L-kynurenine:2-oxoglutarate amino-

Fig. 1. *Activities of tyrosine aminotransferase and tryptophan aminotransferase titrated against anti-(tyrosine aminotransferase) serum*

Procedures for the preparation of liver extracts, assay of enzymes and immunoprecipitation of tyrosine aminotransferase are described in the Materials and methods section. Results are means ±S.E.M. for three independent observations. ○, Enzyme activity in the presence of antiserum; ●, enzyme activity in the presence of control serum.

Table 1. *Effects of anti-(tyrosine aminotransferase) serum on the induction of tyrosine aminotransferase and tryptophan*

Procedures for treatment of animals, preparation of liver extracts, assay of enzymes and immunoprecipitation of tyrosine aminotransferase are described in the Materials and methods section. The results are means ±S.E.M. from three independent observations. Means that are significantly different by Student's *t* test from those for fed control rats are shown by: *$P<0.05$; **$P<0.01$. Other differences are not significant.

	Tyrosine aminotransferase activity (nmol/min per g wet wt. of liver)		Tryptophan aminotransferase activity (nmol/min per g wet wt. of liver)	
	No antiserum	+antiserum	No antiserum	+antiserum
Fed control	1029±66	113±10	51.9±5.4	17.8±0.2
48h-starved	1549±195	121±15	60.9±0.5	30.2±3.8*
Tryptophan-treated	4073±575**	267±68	241.7±64.5*	26.5±3.6
Triamcinolone-treated	4646±664**	203±32	255.1±55.1*	26.8±4.6

Vol. 220

Paragraphs
Sentences
Words

Application Examples

- Documents
- Logs
- Source code
- E-mail archives
- Webpage, blog, wiki, etc.
- Twitter feed, Facebook news feed
- Digital-library
- Patient records
- Speeches
- Poems

- Various domains:
  - Legal matters
  - Intelligence analysis
  - Healthcare
  - Software engineering
  - etc.

Text Visualization

- Transforming text into spatial representation to reveal:
  - Thematic patterns
    - „What are the main topics of a text?“
    - „What events can be tracked over time within a text stream?“
  - Relationship between documents
    - „Which documents contain text about X?“
    - „Are there other documents similar to those I already have?“

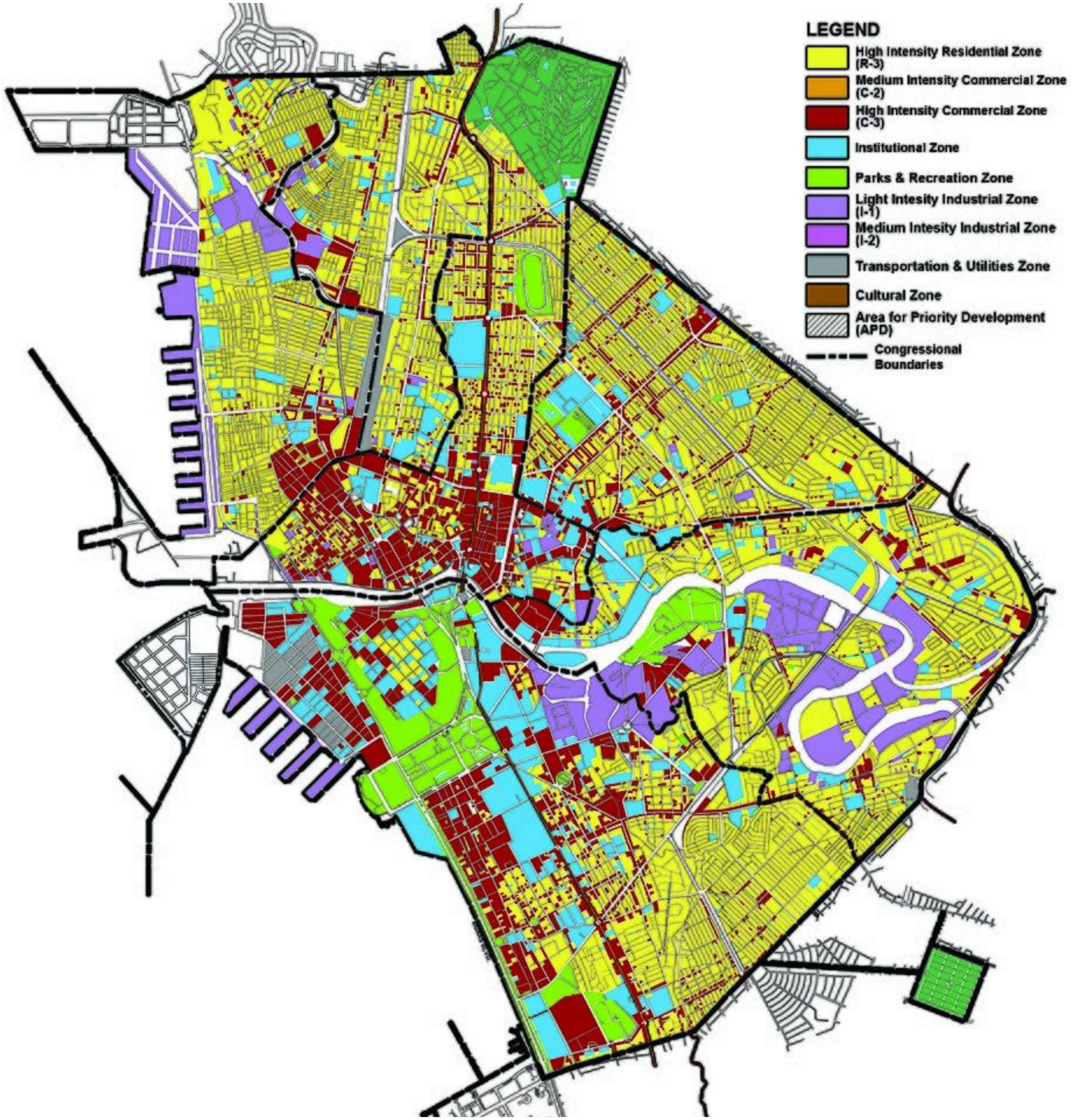
GEOGRAPHICAL DATA

Mixing spatial and abstract data

Geographical Data

- We have geometry
  - Points of interest
  - Polylines, curves - streets, rivers
  - Areas - Country, district, parcel
- To the geometry are attached abstract data
  - Text - name of street or district
  - Population in the area
  - Etc.
- Geographical systems allow us to query both geometrical and abstract data
  - Show areas in '*High intensity residential zone*' that are closer than *5 km* to a river.

Example of Geographical Data



Color-coded zones in Manila city
https://en.wikipedia.org/wiki/Geography_of_Manila

TIME VARYING DATA

Data changes in time

Time Varying Data

- The values of attributes are changing in time
- Topology of grid or network can change in time for spatial fields and networks
- It is better if you think about time varying data as about several datasets
  - Where each dataset have time attached to it
- Different attributes may have different granularity (in time)
  - E.g., we measure temperature each 30 seconds, but measure pressure each minute.

Time modeling

- Ordinal domain – only relative relations between events are considered (The meeting will be after lunch)
- Discrete domain – we view time as discrete; we use nearest neighbor interpolation to interpolate values of attributes (Digital clock)
- Continuous domain – we view time as continuous; we use linear interpolation to interpolate values of attributes

TASK CLASSIFICATION

Why are we visualizing the data?

Task types

- We need to think about tasks in abstract terms
  - To separate the task from the domain language
- Examples
  - **Epidemiologists:** *"I need to contrast the prognosis of patients who were hospitalized more than one month after exposure to patients hospitalized within the first week."*
  - **Biologist:** *"I need to see if results for the tissue samples treated with LL-37 match up with the ones without the peptide."*
- If we think about the tasks in abstract terms, they are just one task
  - *"I need to compare values between two groups."*

Tasks: Actions and targets

- We will distinguish between **action** (verb) and **target** (subject).
- “*I need to* ***compare*** ***values between two groups***.”

Tasks: Actions and targets

Actions

- Analyze
  - *Consume:* Discover, Present, Enjoy
  - *Produce:* Annotate, Record, Derive
- Query
  - Identify
  - Compare
  - Overview/Summarize
- Search
  - Lookup, Locate
  - Browse, Explore

Targets

- All data
  - Trends, Outliers, Features
- Attributes
  - *One:* Distribution, Extremes
  - *Many:* Dependency, Correlation, Similarity
- Networks
  - Topology, Path
- Spatial data
  - Shape, Distance/proximity

Tasks: Analyze actions

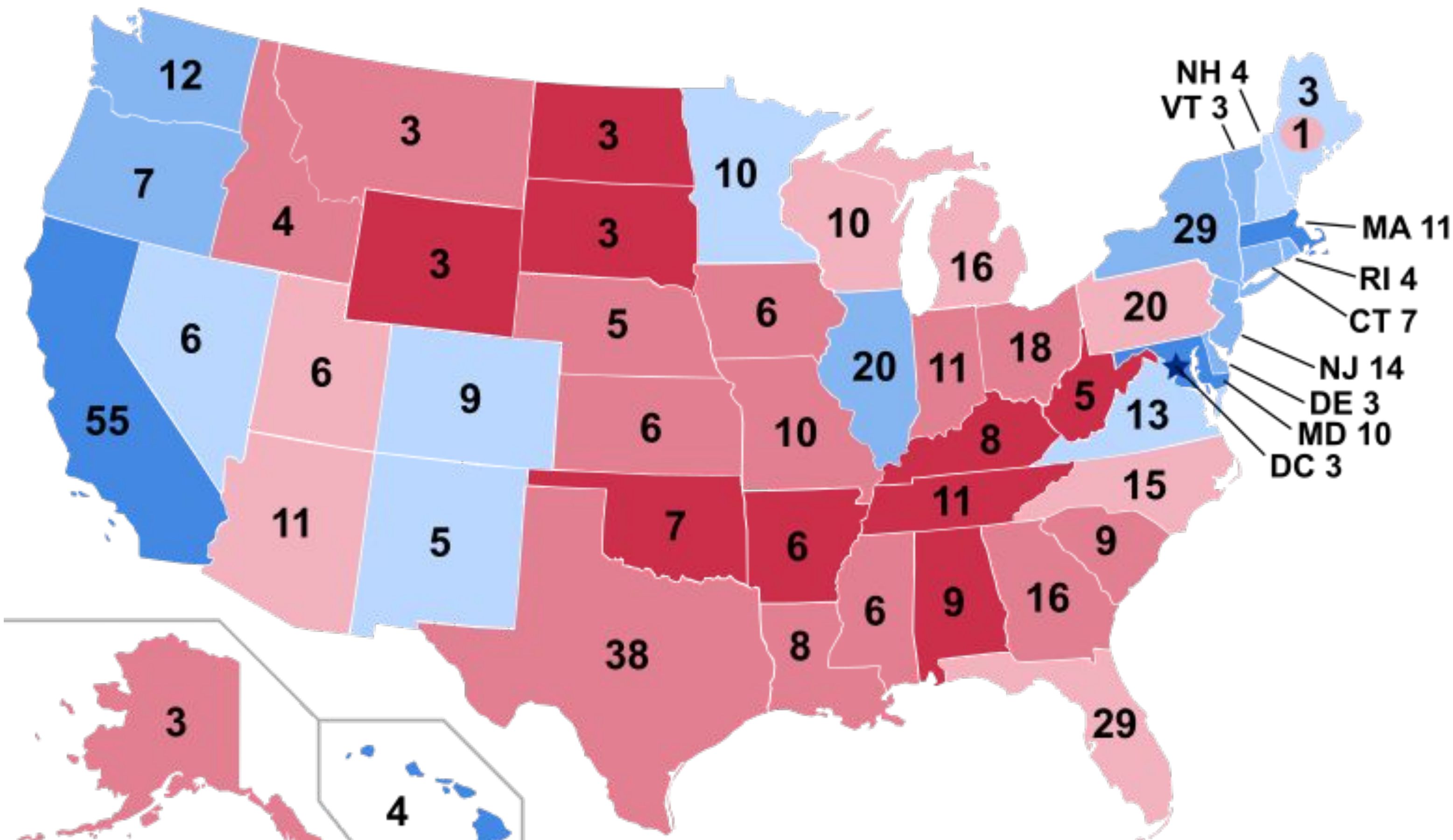
- Consume - the visualization is used to be consumed by certain audience
  - **Discover - the visualizalization is consumed by an expert who can confirm or reject a hypothesis or form a new hypothesis**
  - Present - the visualization is used to transfer knowledge from an expert to the audience
  - Enjoy - the visualization is consumed by a general public just for enjoyment
- Produce - the visualization is used to produce new data
  - Annotate - attach annotations to the data, e.g., attach names to nodes of a network
  - Record - create screenshot from the vis, record interaction with the vis, e.g., for presentation purposes
  - Derive - derive new attributes from the existing ones

Tasks: Query actions

- Identify - The scope is a single target
  - We need to obtain characteristics of the target from the visual representation or
- Compare - The scope is multiple targets
  - We want to compare the targets
  - More difficult than identify task
- Overview/Summarize - The scope is all targets
  - We want to provide overview of the dataset

Tasks: Query actions examples

2016 US Presidential elections



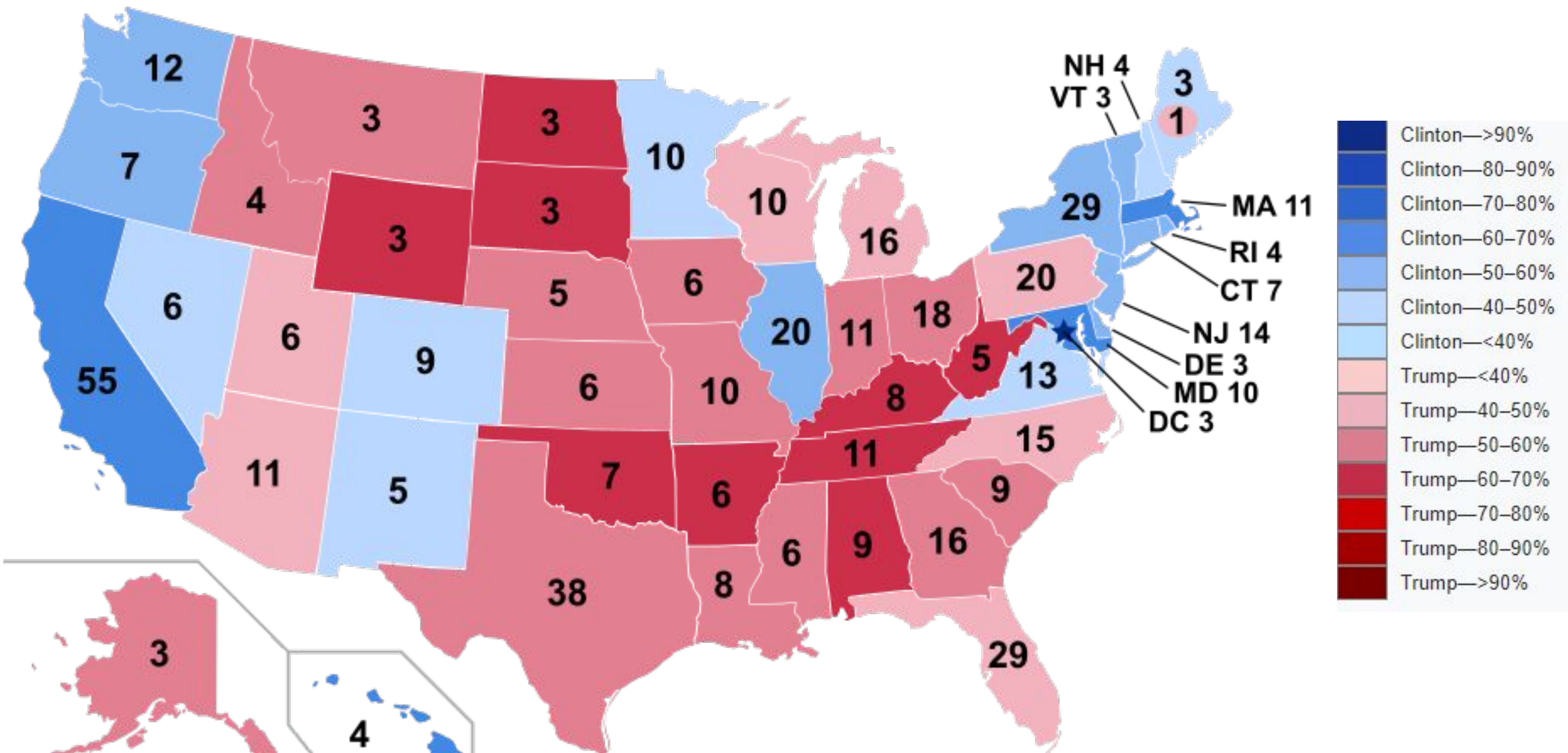
Identify: Who won in California? By what margin?

Target: California

Characteristics: the winner (color hue), the margin of victory (color lightness)

Tasks: Query actions examples

2016 US Presidential elections

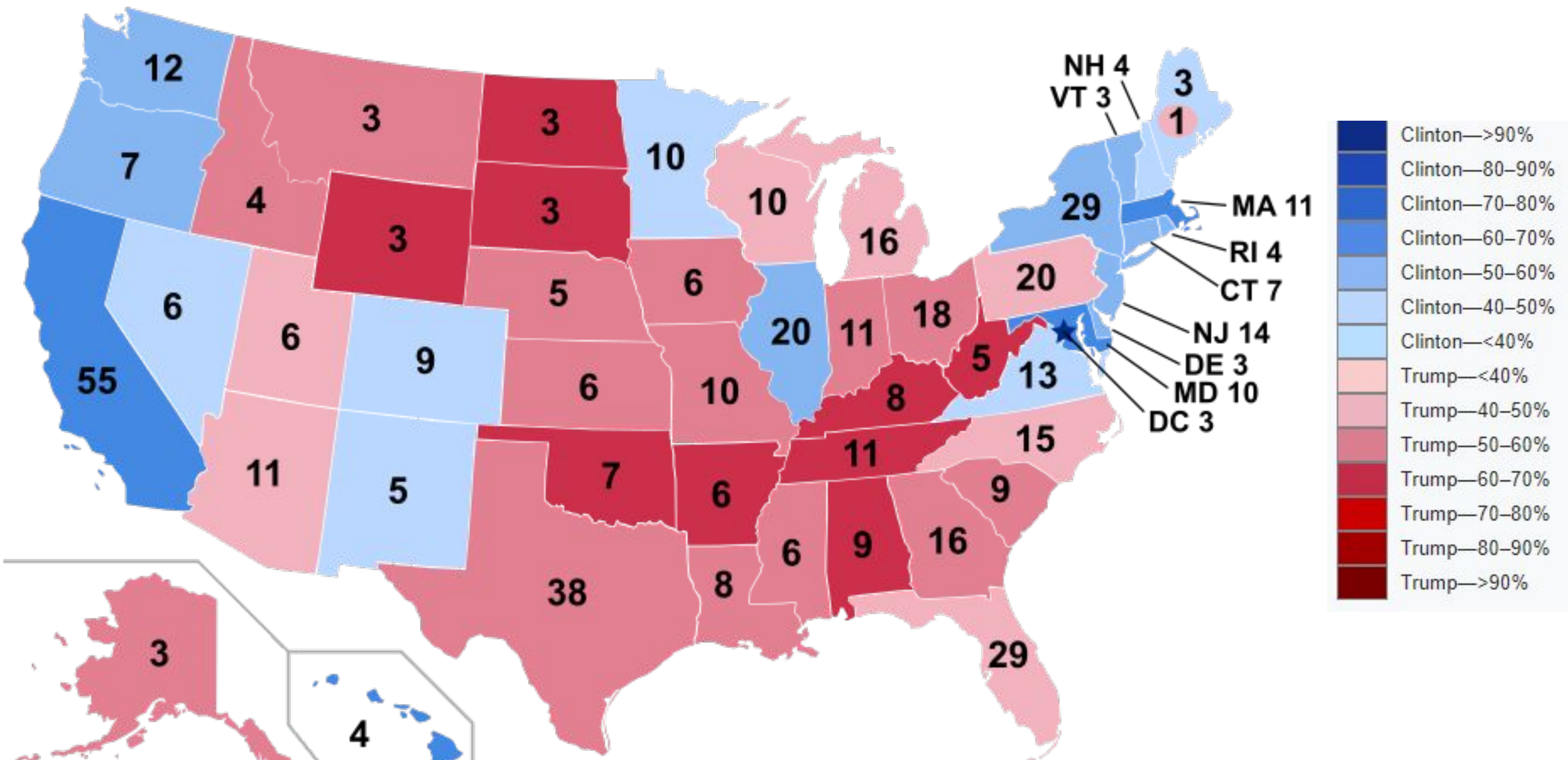


Compare: Is the margin of victory bigger for California or Florida?

Targets: California and Florida   Characteristics: margin of victory (color lightness)

Tasks: Query actions examples

2016 US Presidential elections

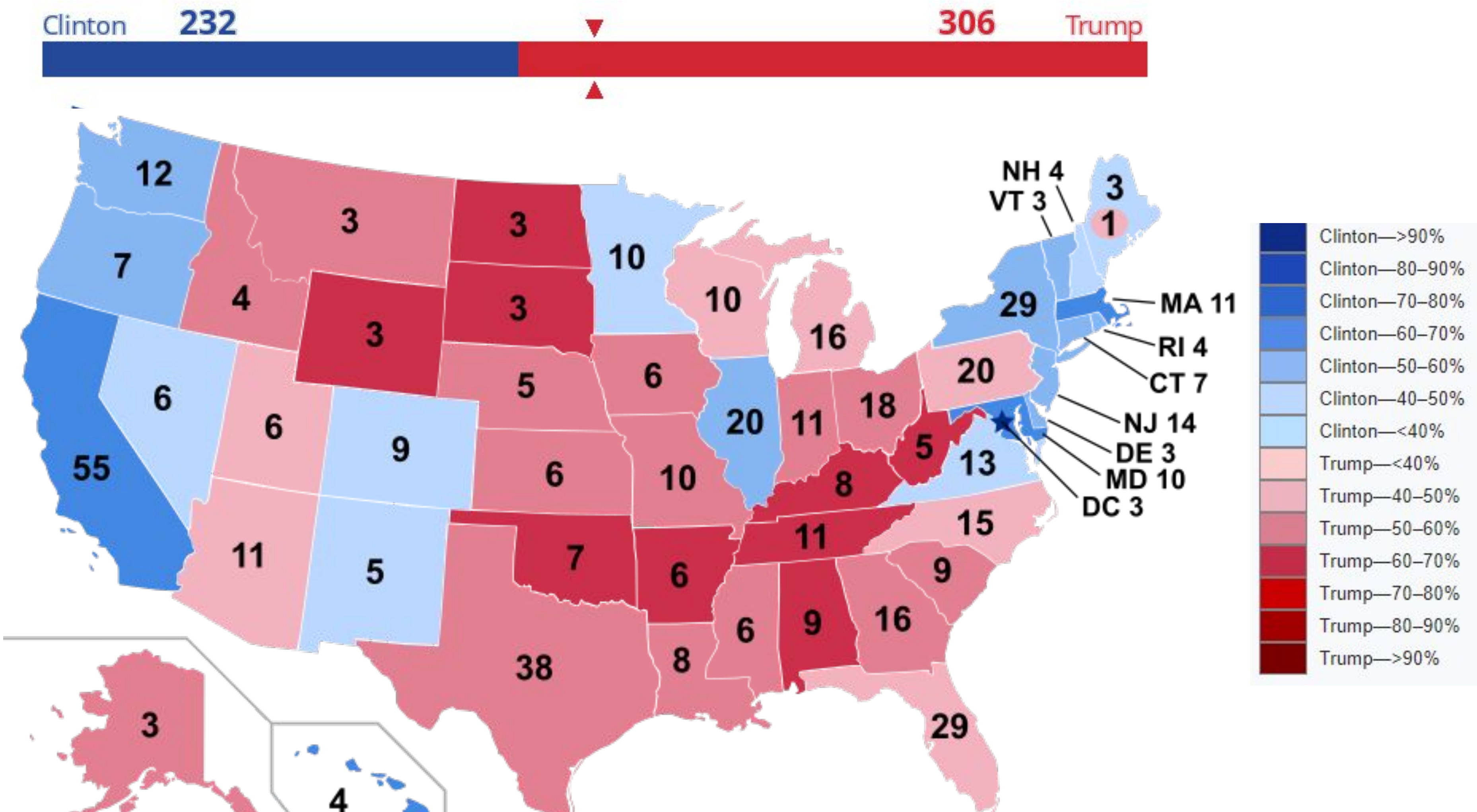


Overview: Who won in most states?

Target: all states Characteristics: the winner (color hue)

Tasks: Query actions examples

2016 US Presidential elections

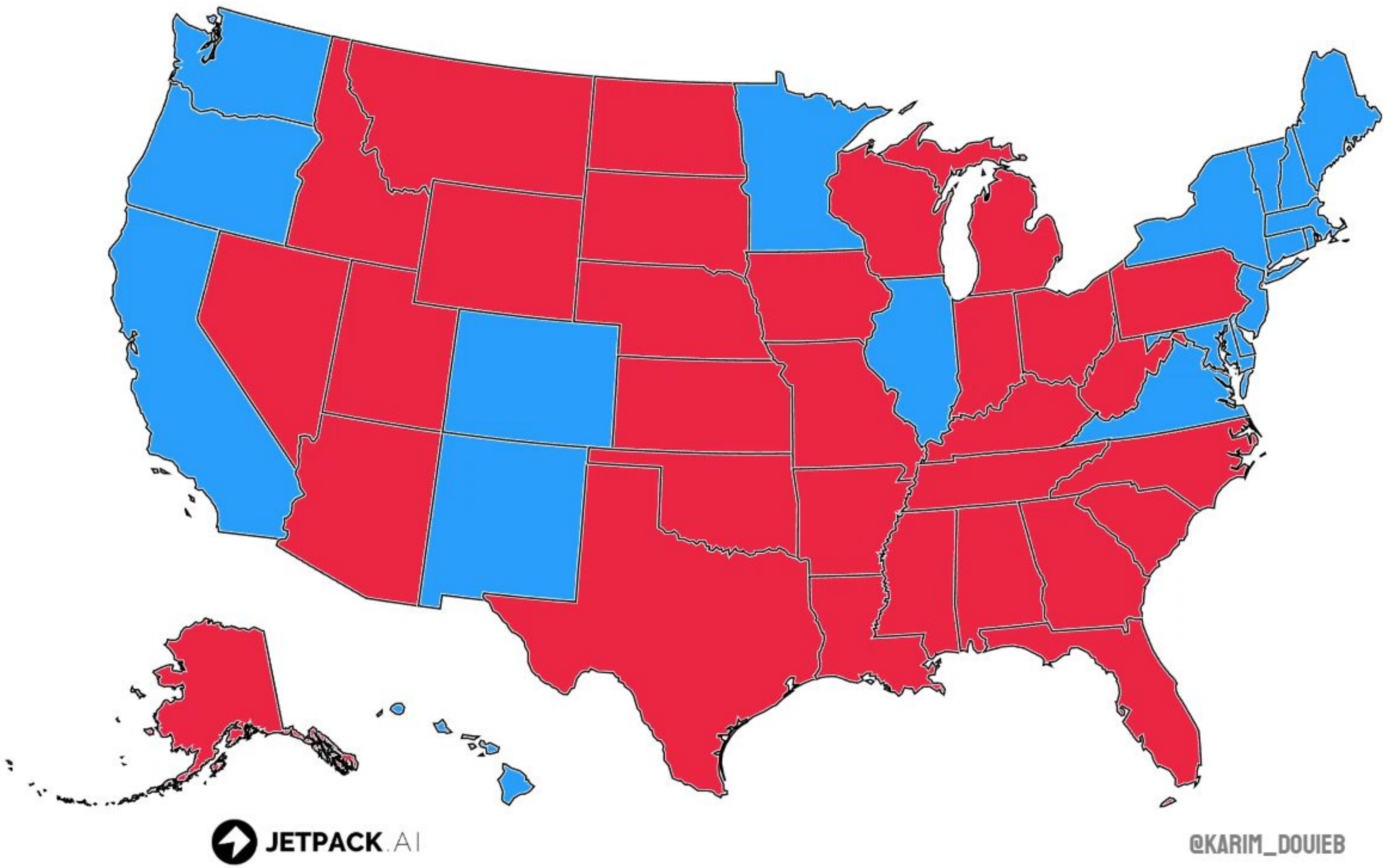


Overview: Who won the elections? By what margin?

We need to aggregate data (members of Electoral College) and compare the values of the aggregated data.

Tasks: Query actions examples

2024 US Presidential elections



Screen capture from: https://observablehq.com/d/3e39dc77f8af714a

Overview: Who won the elections? By what margin?

We need to aggregate data (members of Electoral College) and compare the values of the aggregated data.

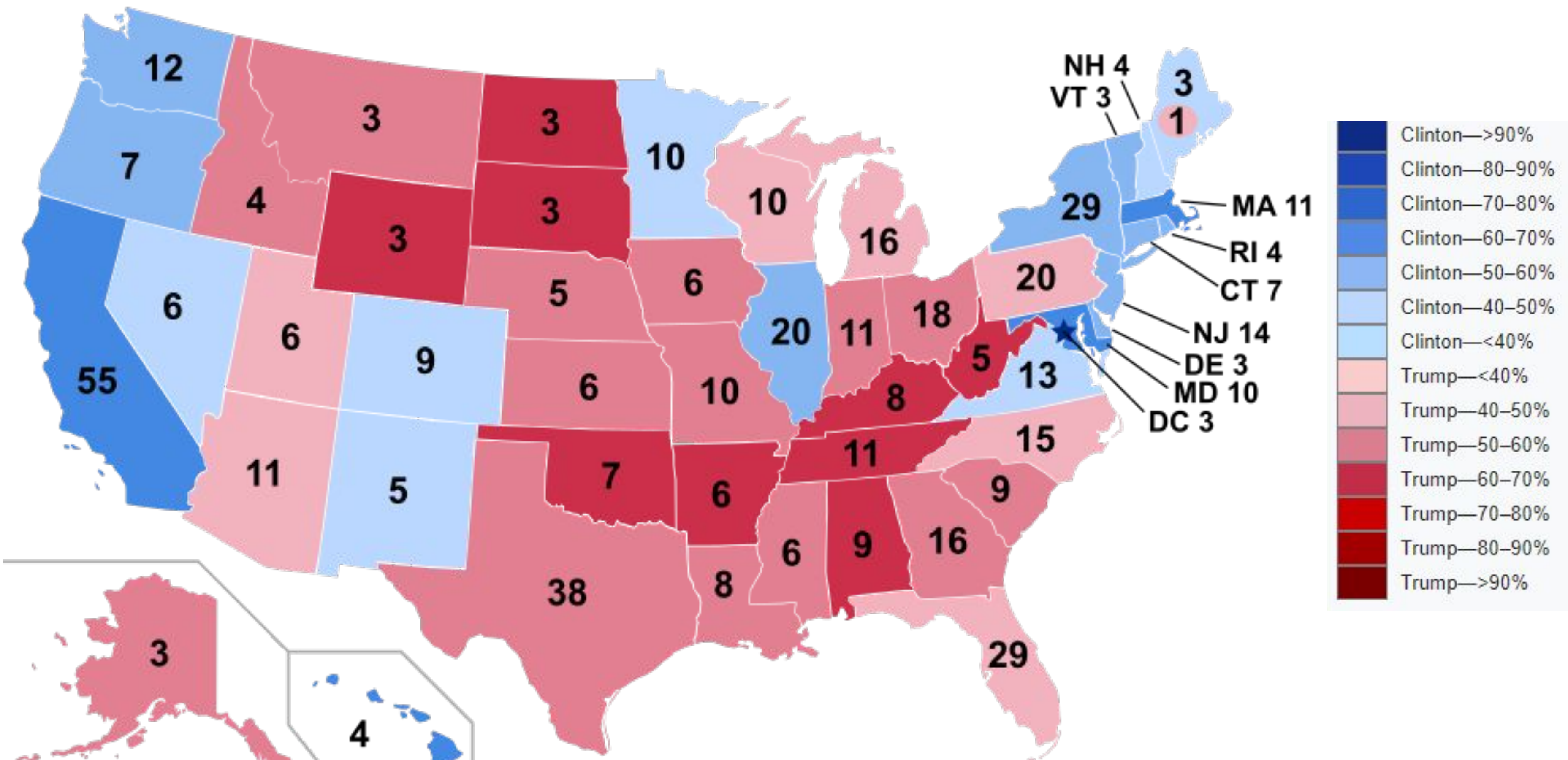
Tasks: Search actions

DCGI

- **Query actions are preceded be one of four search actions:**
- Lookup - Both target and its location are known and we just need to look it up
  - *Who won in California? We know where California is.*
- *Locate* - Target is known, but its location is unknown and we need to locate the target
  - *Who won in Delaware? We don’t know where Delaware is. We need labels of the states to find it (sequential search) or list of the states sorted by alphabet to select Delaware.*
- *Browse* - Target is unknown, but its characteristics are known and we need to find what target/s has/have the characteristics
  - *In what states Clinton won by a big margin? We don’t know the states just its characteristics. We need to find the states.*
- *Explore* - Both target and characteristics are unknown and we explore the dataset to find something interesting.

Tasks: Search actions examples

2016 US Presidential elections



Search: Who won in California? By what margin?

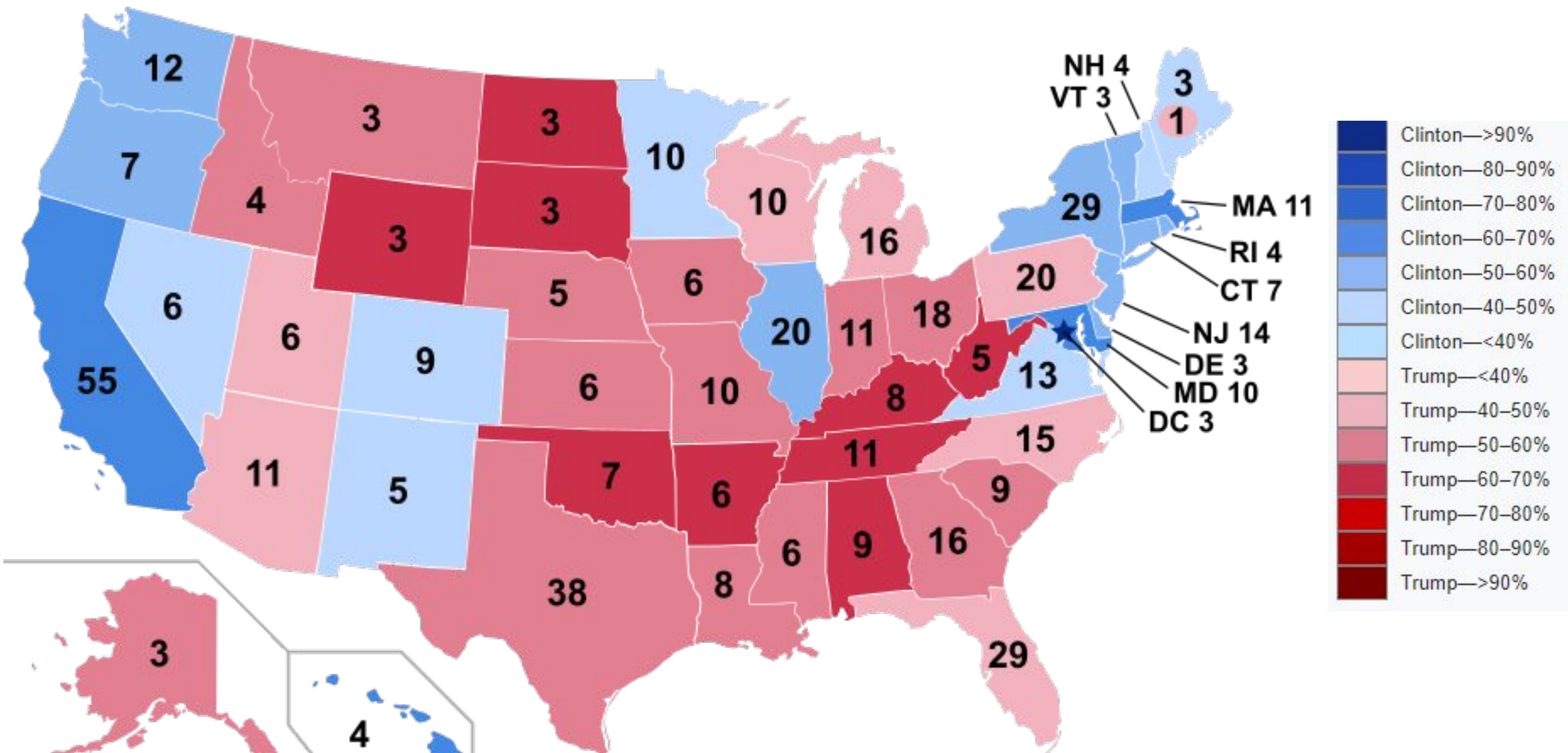
Target: California Characteristics: the winner (color hue), the margin of victory (color lightness)

Tasks: Search actions

- **Query actions are preceded be one of four search actions:**
- Lookup - Both target and its location are known and we just need to look it up
  - *Who won in California? We know where California is.*
- *Locate* - Target is known, but its location is unknown and we need to locate the target
  - *Who won in Delaware? We don't know where Delaware is. We need labels of the states to find it (sequential search) or list of the states sorted by alphabet to select Delaware.*
- *Browse* - Target is unknown, but its characteristics are known and we need to find what target/s has/have the characteristics
  - *In what states Clinton won by a big margin? We don't know the states just its characteristics. We need to find the states.*
- *Explore* - Both target and characteristics are unknown and we explore the dataset to find something interesting.

Tasks: Search actions examples

2016 US Presidential elections



Search: Who won in Delaware? By what margin?

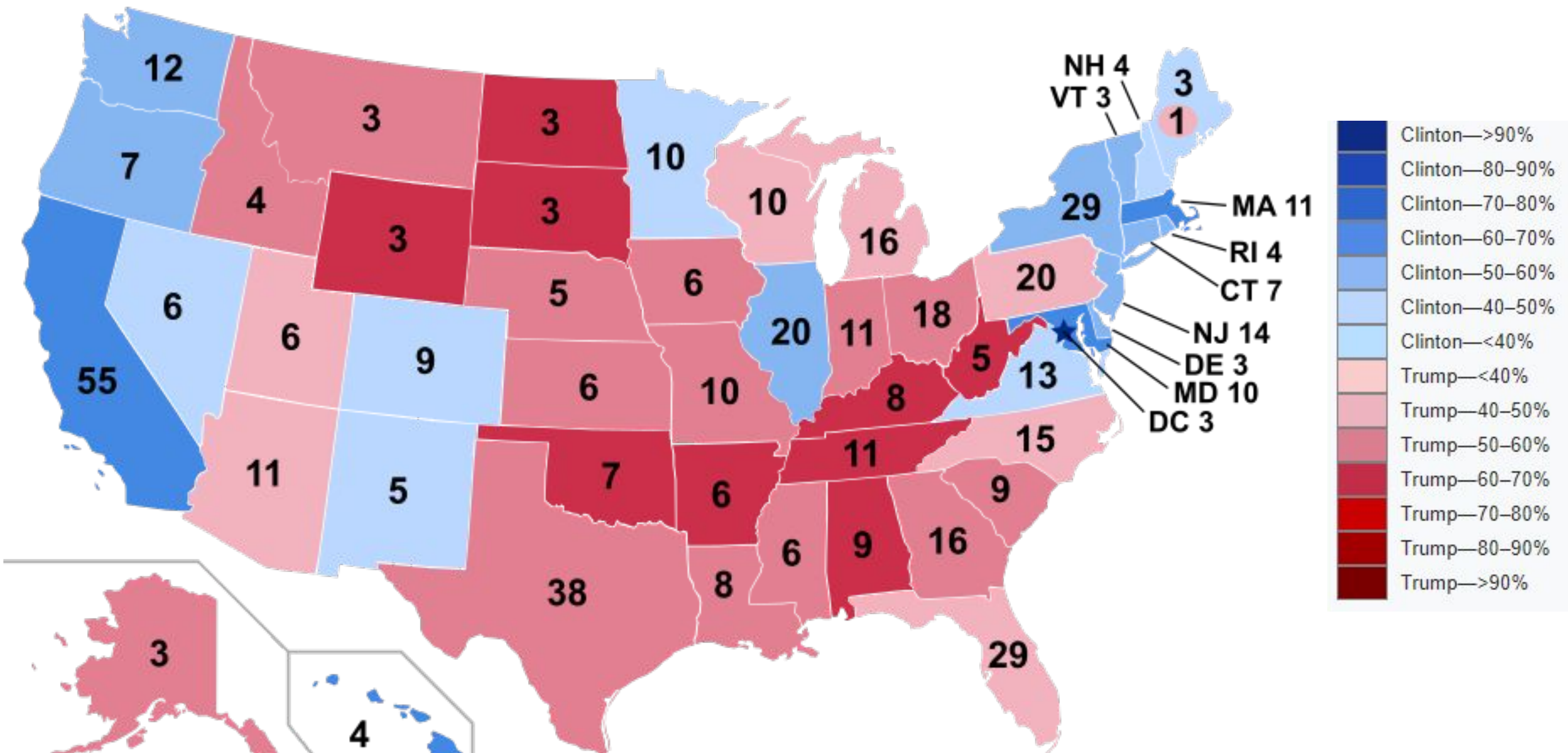
Target: Delaware Characteristics: the winner (color hue), the margin of victory (color lightness)

Tasks: Search actions

- **Query actions are preceded be one of four search actions:**
- Lookup - Both target and its location are known and we just need to look it up
  - *Who won in California? We know where California is.*
- *Locate* - Target is known, but its location is unknown and we need to locate the target
  - *Who won in Delaware? We don't know where Delaware is. We need labels of the states to find it (sequential search) or list of the states sorted by alphabet to select Delaware.*
- *Browse* - Target is unknown, but its characteristics are known and we need to find what target/s has/have the characteristics
  - *In what states Clinton won by a big margin? We don't know the states just their characteristics. We need to find the states.*
- *Explore* - Both target and characteristics are unknown and we explore the dataset to find something interesting.

Tasks: Search actions examples

2016 US Presidential elections



Search: In what states Clinton won by a big margin?

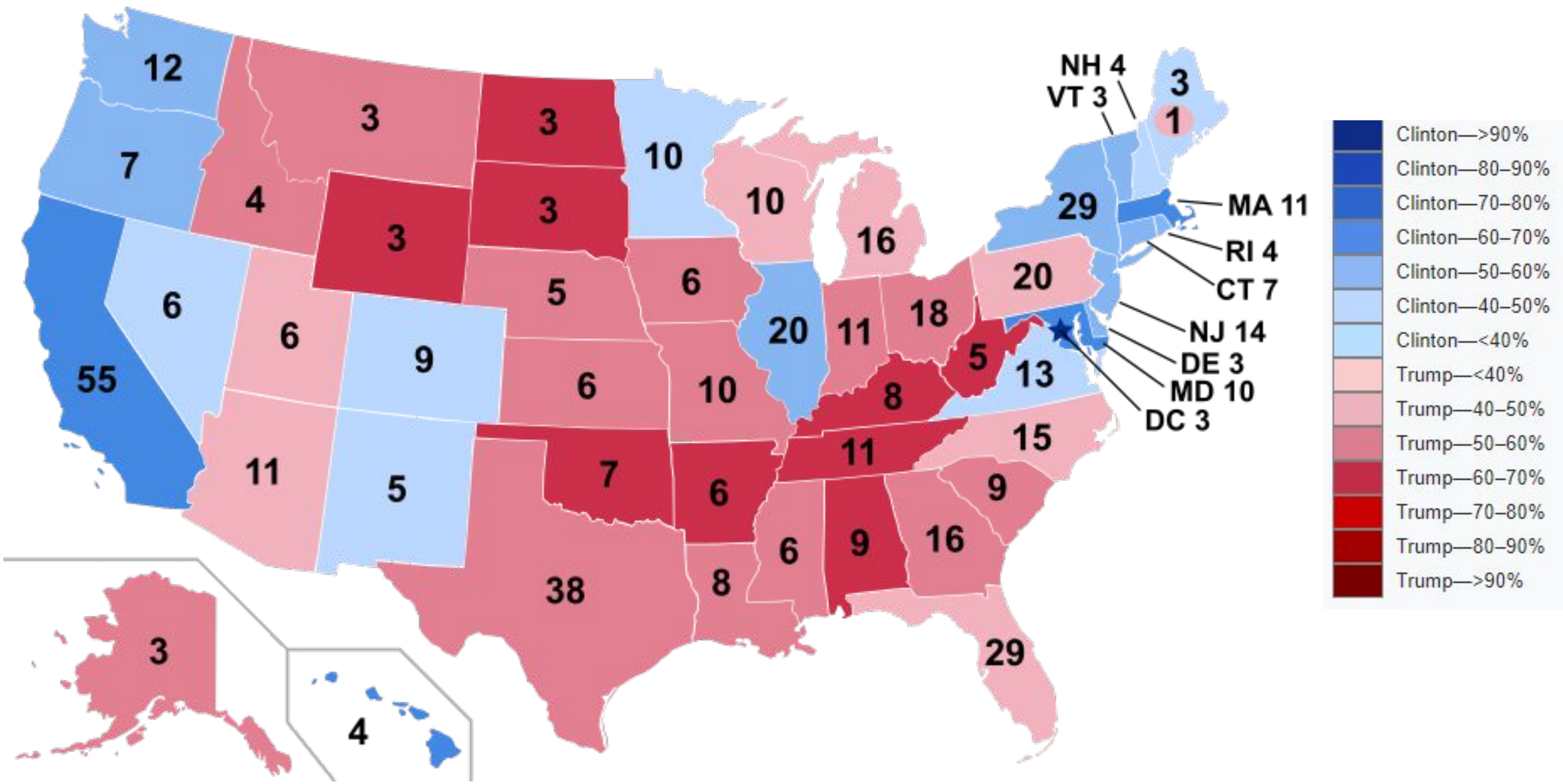
Target: the states   Characteristics: Clinton won by a big margin

Tasks: Search actions

- **Query actions are preceded be one of four search actions:**
- Lookup - Both target and its location are known and we just need to look it up
  - *Who won in California? We know where California is.*
- *Locate* - Target is known, but its location is unknown and we need to locate the target
  - *Who won in Delaware? We don't know where Delaware is. We need labels of the states to find it (sequential search) or alphabetically sorted list of the states to select (and highlight) Delaware.*
- *Browse* - Target is unknown, but its characteristics are known and we need to find what target/s has/have the characteristics
  - *In what states Clinton won by a big margin? We don't know the states just their characteristics. We need to find the states.*
- *Explore* - Both target and characteristics are unknown and we explore the dataset to find something interesting.

Tasks: Search actions examples

2016 US Presidential elections



Tasks: Actions and targets

Actions

- Analyze
  - *Consume:* Discover, Present, Enjoy
  - *Produce:* Annotate, Record, Derive
- Query
  - Identify
  - Compare
  - Overview/Summarize
- Search
  - Lookup, Locate
  - Browse, Explore

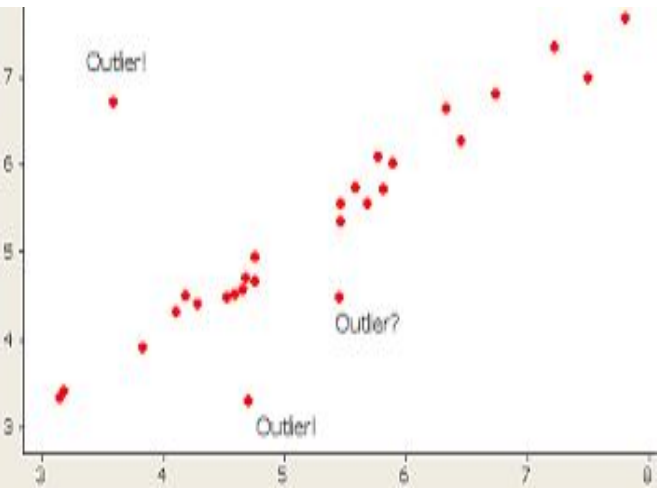
Targets

- All data
  - Trends, Outliers, Features
- Attributes
  - *One:* Distribution, Extremes
  - *Many:* Dependency, Correlation, Similarity
- Networks
  - Topology, Path
- Spatial data
  - Shape, Distance/proximity

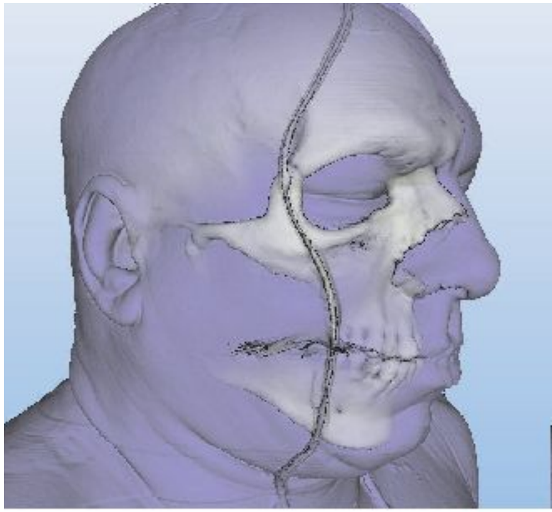
Tasks: Targets

- **All data** - these targets are relevant to all kind of data
  - *Trends* - characterization of a pattern in a data, e.g., increase, decrease, peak, etc.
  - *Outliers* - data that does not fit well into the trend pattern
  - *Features* - certain characteristic of the data, typically domain specific (here bones close to the skin)

Trends



Outliers



Features

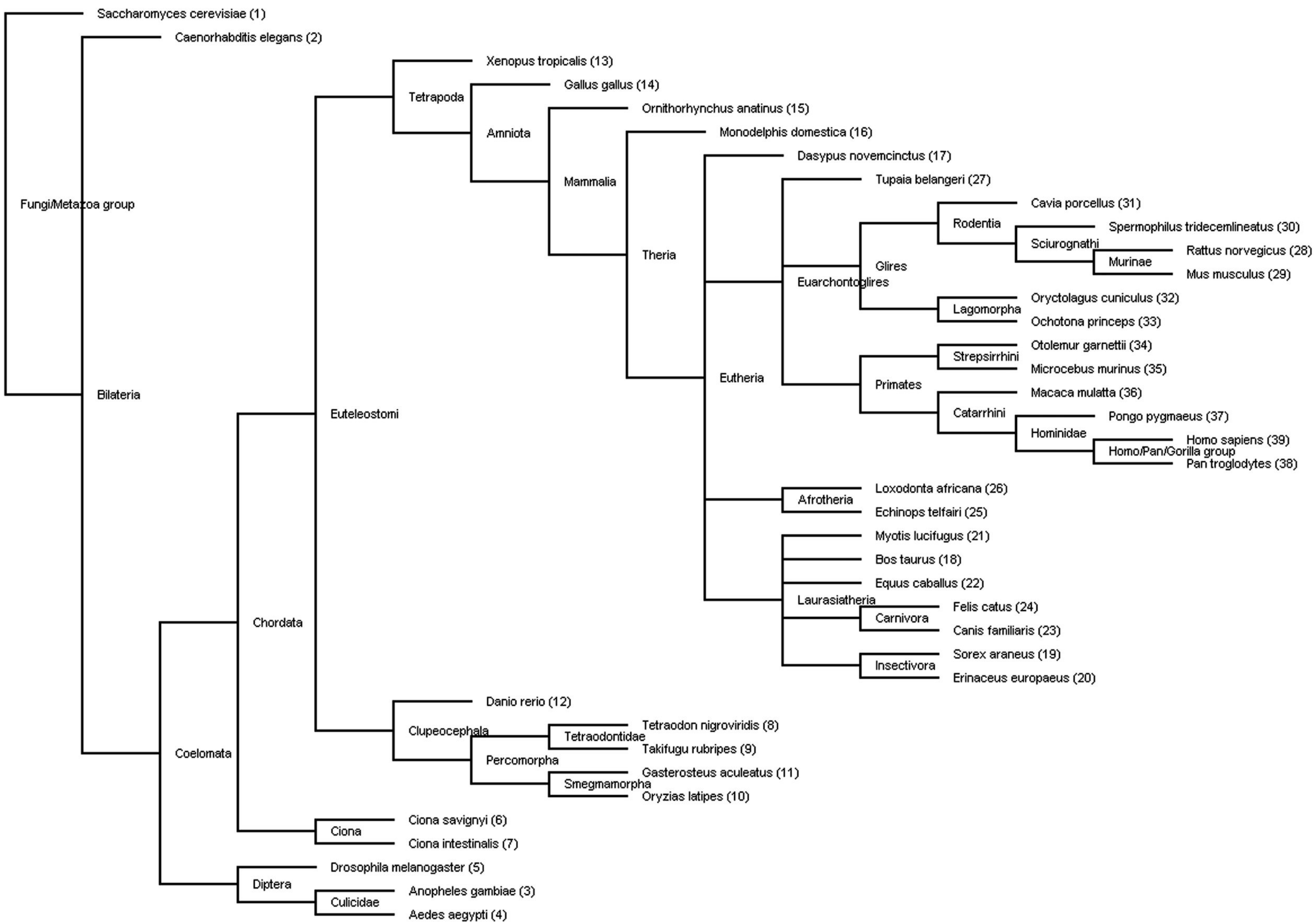
Tasks: Targets

- **Attributes** - these targets are related to the attributes of the datasets
  - **One attribute**
    - *Distribution* - distribution of data for given attribute
    - *Extremes* - minimum and maximum of the attribute
  - **Many attributes**
    - *Dependency* - values of one attribute are dependent on different attribute(s), e.g., the derived attributes
    - *Correlation* - there is tendency for values of one attribute to be tied to another attribute
    - *Similarity* - quantitative measurement calculated for two attributes based on all their values

Tasks: Targets

- **Networks** - targets that may be of interest in networks
  - *Topology* - structure of edges between the nodes of the network
  - *Path* - connection between two nodes of network
- **Spatial data** - targets that may be of interest for spatial data (Fields, Geometries)
  - *Shape* - understanding and comparing the geometric shape
  - *Distance/proximity* - how far/close are certain geometric shapes

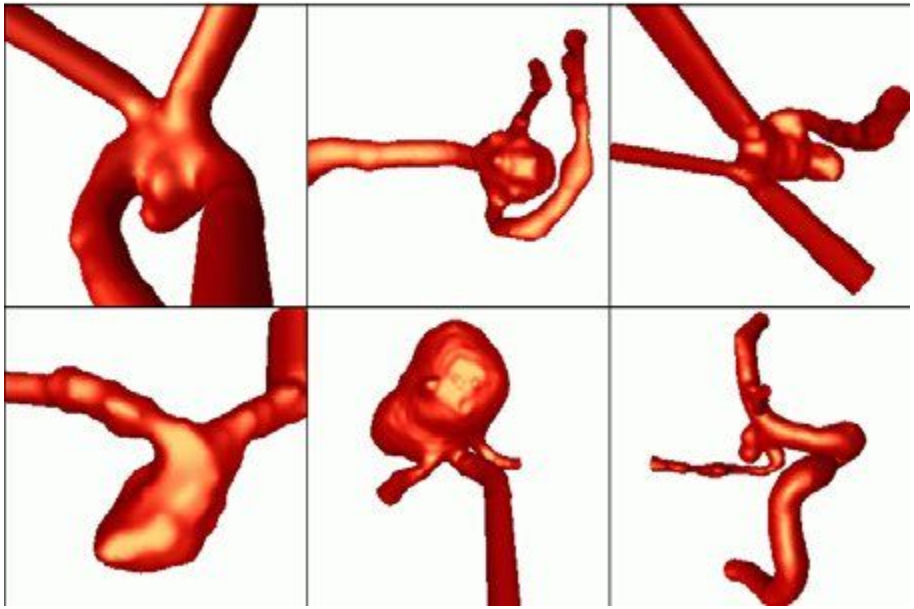
Example: Taxonomy of species



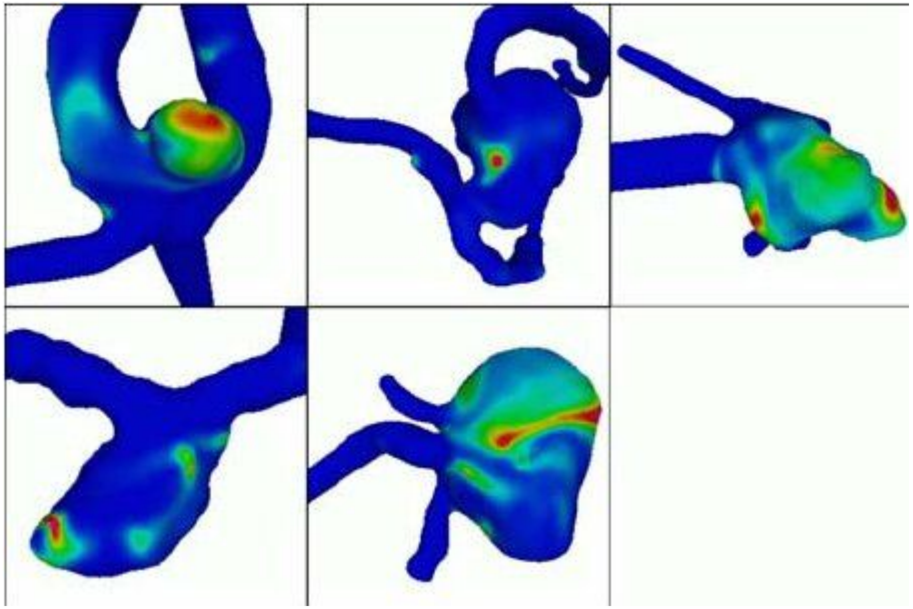
Tasks: Targets

- **Networks** - targets that may be of interest in networks
  - *Topology* - structure of edges between the nodes of the network
  - *Path* - connection between two nodes of network
- **Spatial data** - targets that may be of interest for spatial data (Fields, Geometries)
  - *Shape* - understanding and comparing the geometric shape
  - *Distance/proximity* - how far/close are certain geometric shapes

Example: Brain aneurysm



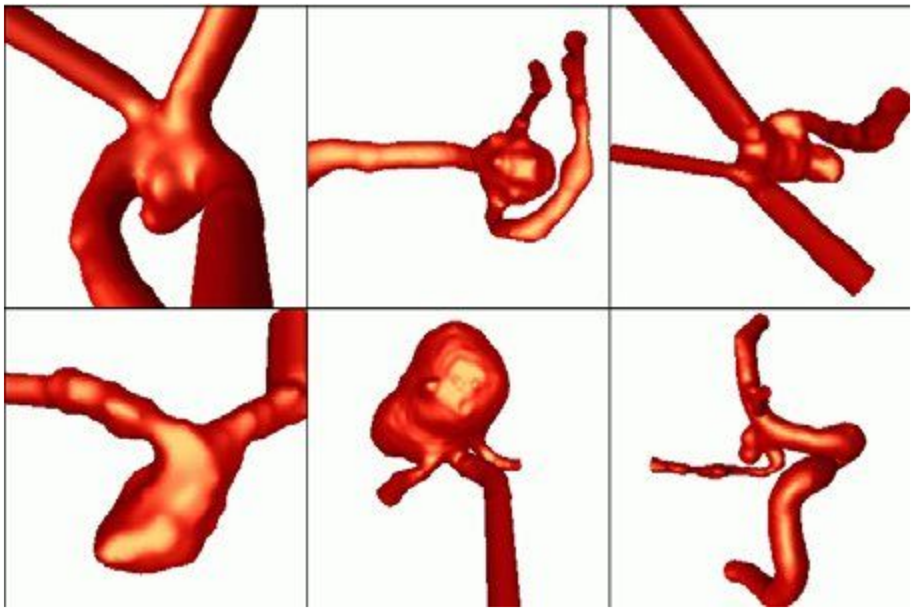
Aneurysms on veins in brain



Velocity of the blood - higher velocity means higher pressure on the vein

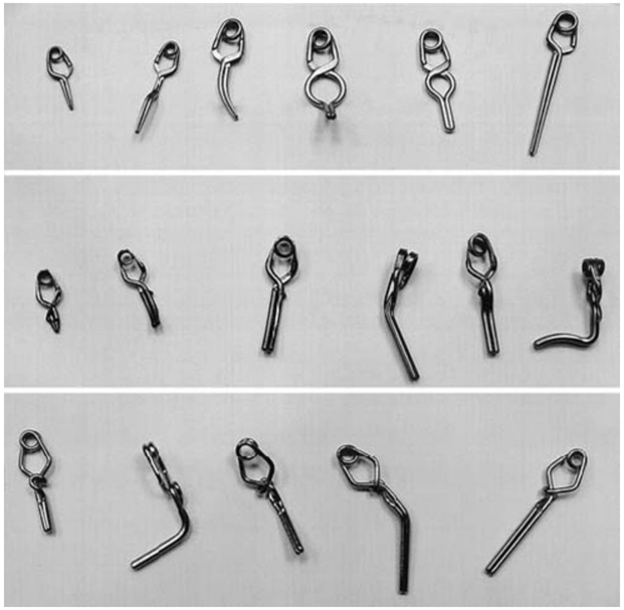
Source of images: CEBRAL, Juan R.; HERNÁNDEZ, Mónica; FRANGI, Alejandro F. Computational analysis of blood flow dynamics in cerebral aneurysms from CTA and 3D rotational angiography image data. In: *International congress on computational bioengineering*. 2003. p. 191-198.

Example: Brain aneurysm



Aneurysms on veins in brain

Source: CEBRAL, Juan R.; HERNÁNDEZ, Mónica; FRANGI, Alejandro F. Computational analysis of blood flow dynamics in cerebral aneurysms from CTA and 3D rotational angiography image data. In: *International congress on computational bioengineering*. 2003. p. 191-198.



Clips used to prevent blood flow into the aneurysm

Source: Louw, D. F., Kaibara, T., & Sutherland, G. R. Aneurysm clips, Journal of Neurosurgery, 98(3), 638-641, 2003.

How we allow users to accomplish tasks?

- Encoding of attributes
  - Mapping data on graphical elements
  - Ordering, alignment, and separation of the graphical elements
- Manipulation with the data
  - Change of parameters of visualization
  - Selection
  - Navigation in the data
- Reduction of the data
  - Filtering
  - Aggregation
- Organization of data views
  - Juxtaposition, Superimposition, Embedding
  - Distribution of data between the views
- *All these topics are covered in the following lectures*

Thank you for your attention